

Course code: CE 353

Course Name: Traffic Engineering

Pre-requisite: None

Course Objectives:

At the end of this course, a student will be able to:

1. Identify and visualize the extent of road congestion, delays, and emissions
2. Estimate the capacity of various road facilities
3. Model road accidents using statistical concepts and models.
4. Design a traffic signal for optimal traffic control
5. Use Intelligent Transportation System for traffic management

Course Content:

UNIT I: Time Space Diagram, Traffic Parameter Measurements. Sensing and Measuring Traffic Parameters (7 Lectures)

UNIT II: Traffic Engineering Models (17 Lectures)

Fundamental Diagram, Shockwave Analysis, Arrival Departure Curves, Macroscopic Models, Microscopic Models, Traffic Simulation, Cell Transmission Model, Gap Acceptance, Queuing Theory and Applications

UNIT III: Statistical Methods for Traffic Engineering (6 Lectures)

Probability Theory, Probability Distributions, Mathematical Statistics, Hypothesis Testing and Statistical Inference, Accident Analysis

UNIT IV: Capacity Analysis and Signal Design (5 Lectures)

Passenger Car Unit, Road Capacity, Level of Service (LOS), Highway Performance Measures, Delay Studies, Design of Signals

UNIT V: ITS and Traffic Operations on the Freeway (4 Lectures)

Travel Time Reliability, Intelligent Transportation System (ITS) and Categories, Sensors for ITS, Freeway Ramp and Variable Speed Limit, Incident Detection

Text Books:

- Chakroborty, P., & Das, A. (2017). Principles of transportation engineering. PHI Learning Pvt. Ltd. Second Edition.
- Roger P. Roess; Elena S. Prassas; William R. McShane. (2019) Traffic Engineering. 5th Edition. Pearson: 9780134599717

References:

- Knoop, V. (2021). Introduction to traffic flow theory: An introduction with exercises.
- Daganzo, C., & Daganzo, C. F. (1997). Fundamentals of transportation and traffic operations (Vol. 30). Oxford: Pergamon.
- HCM, Highway Capacity Manual, Transportation Research Board, USA. (2022 edition)
- Indo HCM, Highway Capacity Manual, India (2017 edition)

Assessment:

- Mid-Term
- End-Term
- Quiz
- Assignment